

TheLook E-Commerce Analytics Dashboard

Project Report

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Role: Data Analyst Portfolio Project

Tools Used: SQL, Google BigQuery, Power BI, DAX

Dataset: TheLook E-Commerce Dataset (Google BigQuery Public Dataset)

Project Type: End-to-End Business Intelligence & Data Analytics Project

Date: June 2026

1. Executive Summary:

The objective of this project was to analyze e-commerce sales performance, customer behavior, product profitability, and marketing effectiveness using TheLook E-Commerce dataset available in Google BigQuery.

An end-to-end Business Intelligence solution was developed using SQL, BigQuery, Power BI, and DAX to transform raw transactional data into actionable business insights.

The final deliverable is a four-page interactive dashboard covering:

- Executive Overview
- Customer Analytics
- Product & Profitability Analytics
- Marketing Analytics

The project enables stakeholders to monitor business performance, identify profitable customer segments, evaluate product performance, and assess marketing channel effectiveness.

2. Business Objectives:

The project was designed to answer the following business questions:

Sales Performance

- How much revenue and profit is generated?
- How are sales changing over time?
- What is the average order value?

Customer Analytics

- Who are the customers?
- Which customer segments generate the most revenue?
- What is the repeat customer rate?
- What is the average customer lifetime value?

Product Analytics

- Which categories generate the highest revenue?
- Which categories generate the highest profit?
- Which brands perform best?
- Which categories maintain the highest profit margins?

Marketing Analytics

- Which acquisition channels drive customer growth?
- Which channels generate the most revenue?
- How has channel performance evolved over time?

3. Dataset Overview:

Dataset Source:

Google BigQuery Public Dataset – TheLook E-Commerce

Tables Used:

users: Customer demographic and acquisition information.

orders: Order-level information including order status and fulfillment dates.

order_items: Transaction-level sales records.

products: Product information including category, brand, retail price, and cost.

Dataset Size:

Table	Records
Users	100,000
Orders	125,270
Order Items	181,662
Products	29,120

Analysis Period: 2019 – 2026

4. Data Exploration & Validation:

Data quality checks were performed before analysis.

Duplicate Check: No duplicate records were identified in the primary tables.

Date Validation: Data spans from 2019 to 2026, enabling trend analysis.

Order Status Distribution:

Status	Transactions
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Shipped	54,508
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Complete	45,497
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Processing	36,207
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Cancelled	26,874
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Returned	18,576
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Only completed and shipped orders were included in revenue calculations.

5. Data Modeling:

A Star Schema was implemented in Power BI.

Fact Table: fact_order_items

Dimension Tables

- dim_users
- dim_orders
- dim_products
- dim_date

Relationships

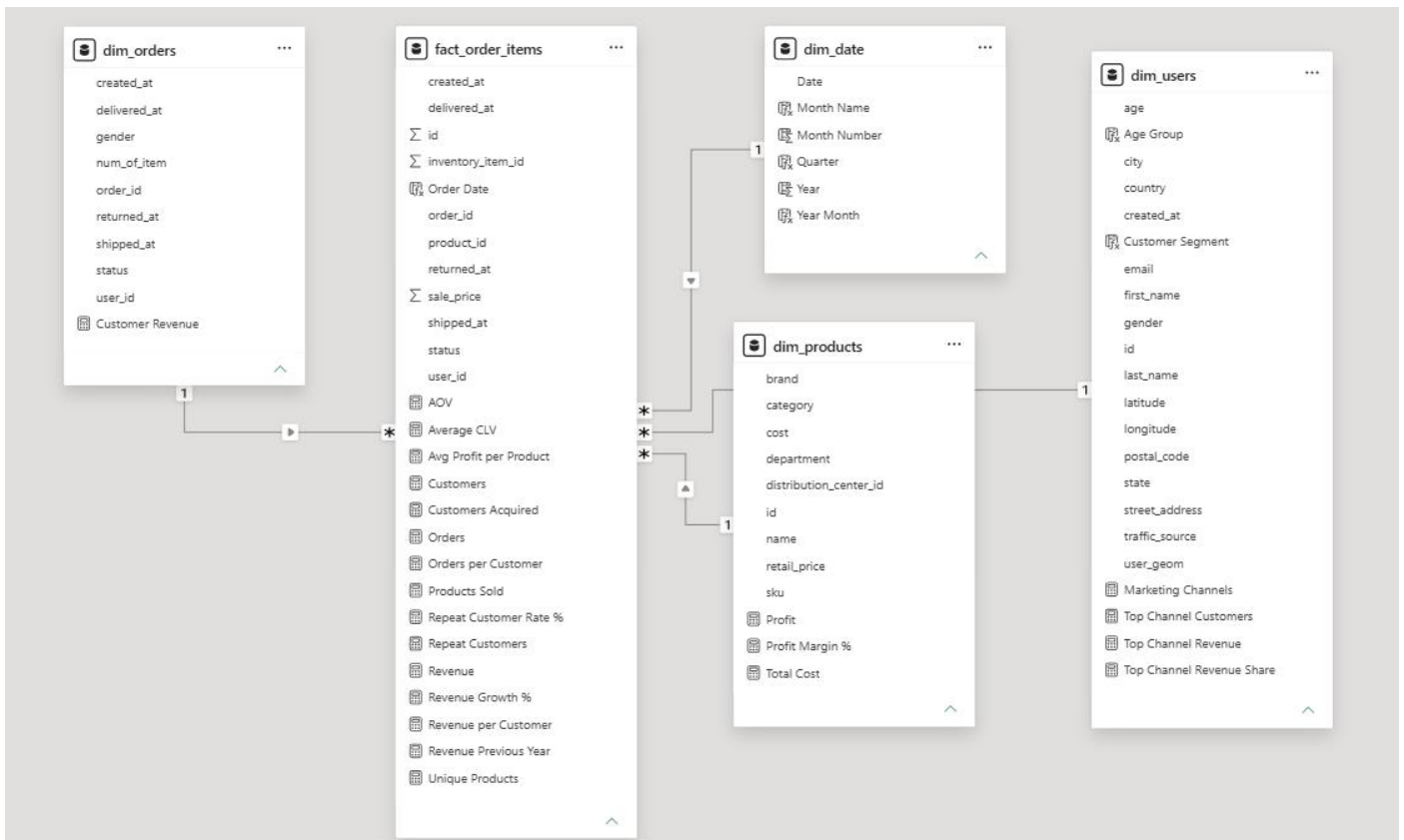
users → order_items

orders → order_items

products → order_items

date → order_items

This model supports efficient reporting and simplifies DAX calculations.



6. Key Metrics Developed:

Revenue: Total sales generated from completed and shipped orders.

Orders: Distinct count of orders.

Customers: Distinct count of customers.

Average Order Value (AOV): Revenue divided by total orders.

Profit: Revenue minus product cost.

Profit Margin %: Profit as a percentage of revenue.

Customer Lifetime Value (CLV): Total revenue generated by each customer throughout their relationship with the business.

Repeat Customer Rate

Percentage of customers placing multiple orders.

7. Dashboard Development:

Page 1: Executive Overview

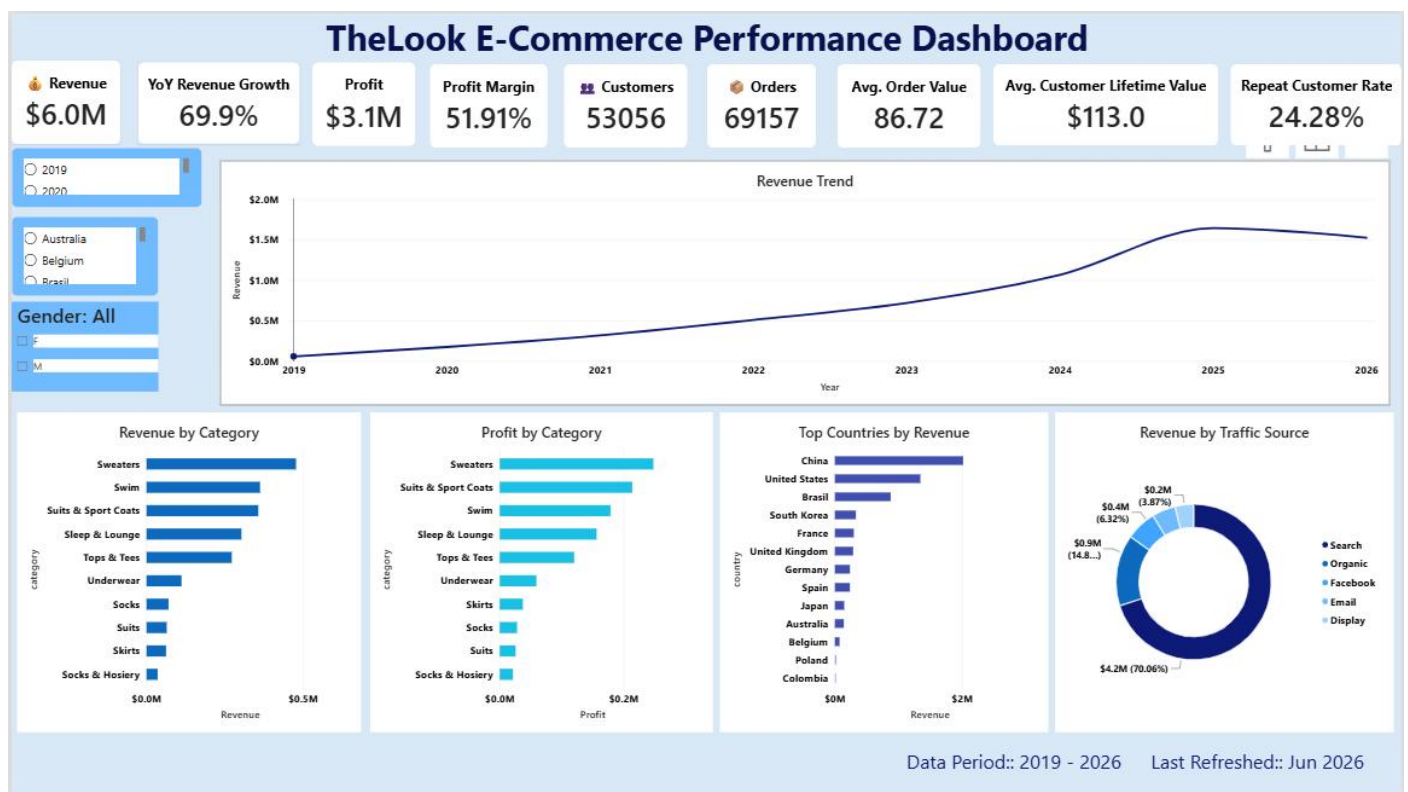
Purpose: Provide a high-level view of business performance.

KPIs Included:

- Revenue
- Profit
- Customers
- Orders
- Average Order Value
- Profit Margin
- Average CLV
- Repeat Customer Rate

Visualizations:

- Revenue Trend
- Revenue by Category
- Profit by Category
- Revenue by Country



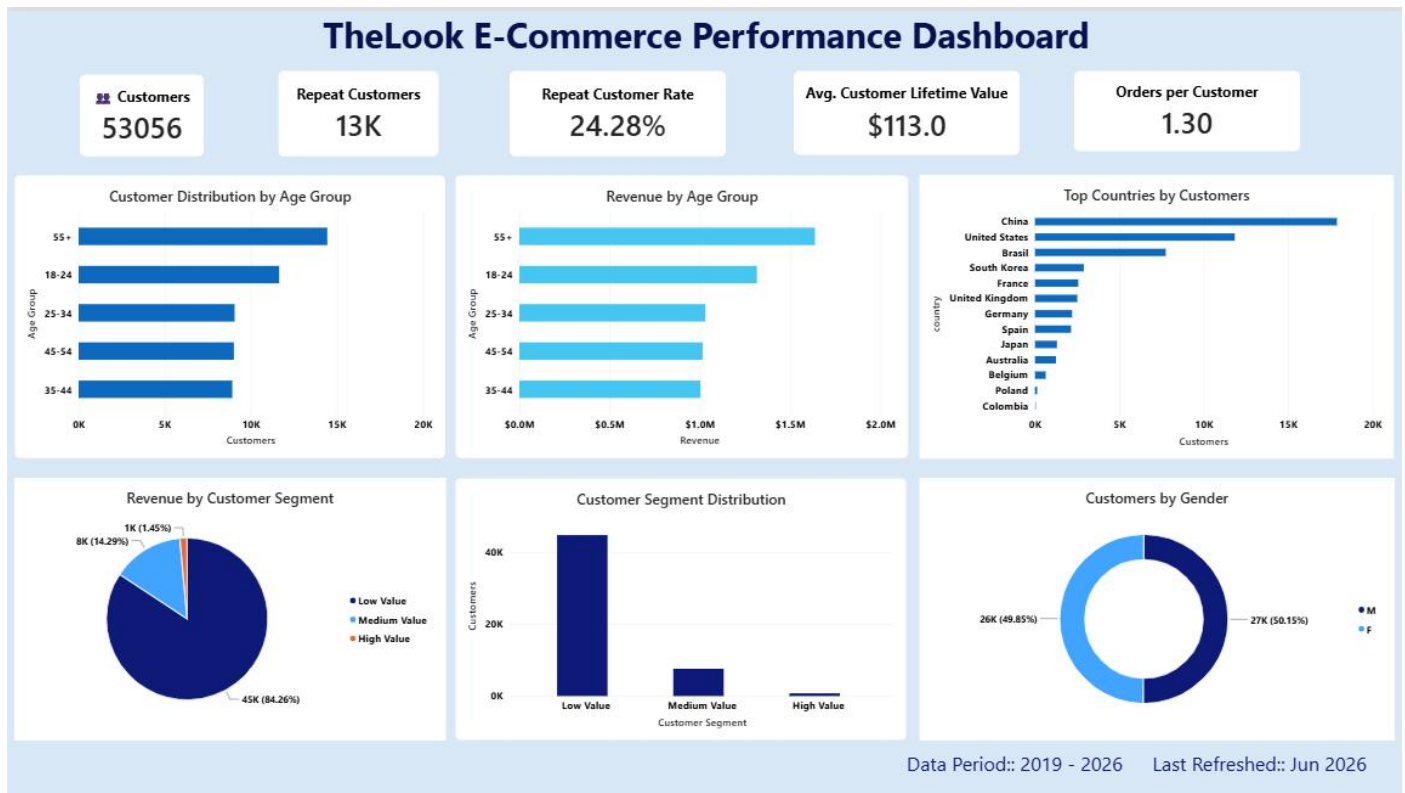
Page 2: Customer Analytics

Purpose: Understand customer demographics and behavior.

Visualizations:

- Customer Distribution by Age Group
- Revenue by Age Group
- Customer Distribution by Gender

- Revenue by Customer Segment
- Customer Segment Distribution
- Top Countries by Customers



Page 3: Product & Profitability Analytics

Purpose: Identify high-performing products and profitable categories.

Visualizations:

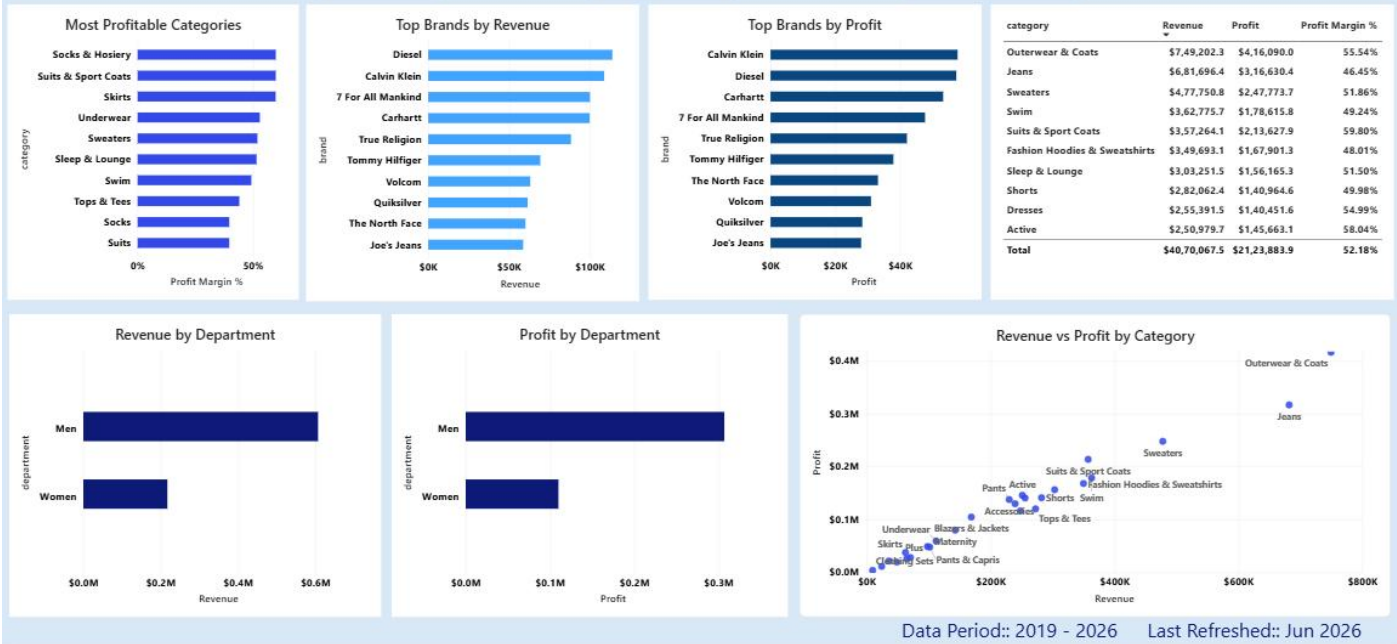
- Most Profitable Categories
- Revenue by Department
- Profit by Department
- Top Brands by Revenue
- Top Brands by Profit
- Revenue vs Profit by Category

TheLook E-Commerce Performance Dashboard

Products Sold
100K

Unique Products
28.109K

Avg Profit per Product
\$31.1

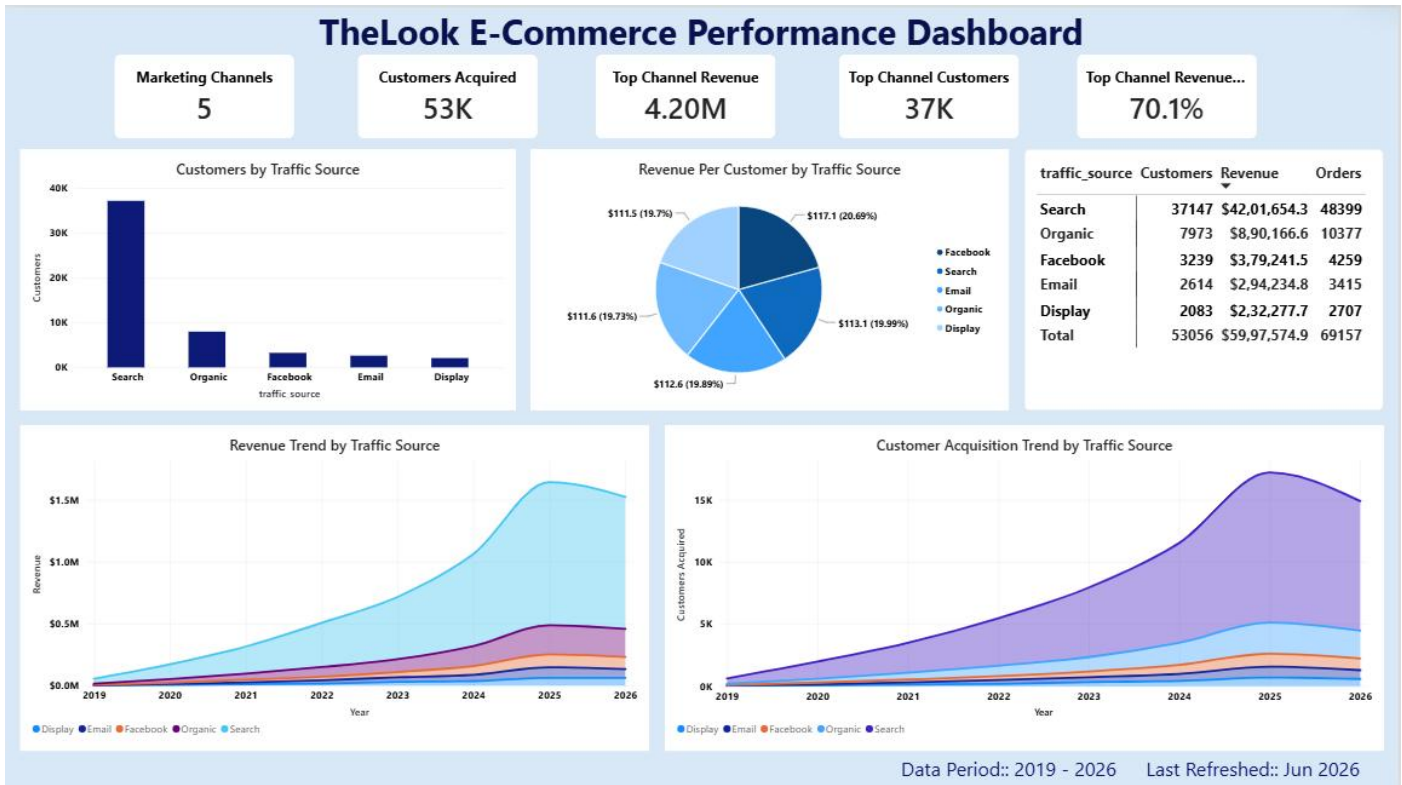


Page 4: Marketing Analytics

Purpose: Evaluate acquisition channel performance.

Visualizations:

- Revenue by Traffic Source
- Customers by Traffic Source
- Revenue Trend by Traffic Source
- Customer Acquisition Trend by Traffic Source
- Channel Performance Matrix



8. Key Findings:

Sales Performance

- Generated approximately \$5.96M in revenue.
- Achieved approximately \$3.09M in profit.
- Maintained a profit margin of approximately 52%.

Customer Insights

- Total customers exceeded 52,000.
- Repeat customer rate was approximately 24%.
- Average customer lifetime value was approximately \$113.

Product Insights

- Outerwear & Coats generated the highest revenue and profit.
- Multiple product categories maintained strong profit margins.
- Profitability varied considerably across categories.

Marketing Insights

- Search was the dominant acquisition channel.
- Search generated approximately 70% of total revenue.
- Organic traffic was the second most effective channel.

Growth Insights

- Revenue increased consistently from 2019 through 2025.
- Customer acquisition followed a similar upward trend.
- Business performance demonstrated sustained growth over time.

9. Business Recommendations:

Customer Strategy

- Increase retention efforts to improve repeat customer rate.
- Develop loyalty programs targeting medium and high-value customers.

Product Strategy

- Prioritize investment in high-margin categories.
- Expand inventory in top-performing product segments.

Marketing Strategy

- Continue investing in Search acquisition channels.
- Optimize Organic channel growth initiatives.
- Evaluate the efficiency of lower-performing channels.

Revenue Growth

- Focus on converting low-value customers into repeat buyers.
- Use customer segmentation to support personalized marketing campaigns.

10. Conclusion:

This project demonstrates the application of SQL, BigQuery, Power BI, DAX, and Business Intelligence principles to solve real-world e-commerce analytics problems.

The developed dashboard provides decision-makers with a comprehensive view of sales performance, customer behavior, product profitability, and marketing effectiveness, enabling data-driven business decisions.